

ICAN: Information Capacity Approximate Network for Estimating Regression Model Confidence - ICPR 2026



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Introduction & Background

Neural Networks are widely used for complex regressions tasks, yet measuring the confidence of a **single** prediction remains a significant challenge.

We approach this problem through the lens of Information Theory, specifically Shannon's capacity theorem, which defines the theoretical maximum data transmission rate, over a communication channel.

It is also known as the **achievable rate** of a communication channel.

The transmitter transmits the signal X , and the receiver receives the distorted noisy signal, Y , when X passes through the channel.

Shannon's capacity is given by: $C = \text{Max}\{I(X;Y)\}$

Where: $I(X;Y)$ is the mutual information between X and Y .

A common communication model is given by: $Y = Hx + v$, Where: H is the communication channel, and v is an additive white gaussian noise (AWGN).

The channel capacity in this case, according to Information Theory, is:

$$C = 0.5 \log_2(1 + \text{SNR}) = 0.5 \log_2(1 + \frac{Y^2}{\mathbb{E}\{v^2\}}) \text{ [bits/sec per channel usage]}$$

Where: $\mathbb{E}\{v^2\}$ is the expectation of v^2 (that is, the variance of the noise), and Y^2 is the energy of Y , the received signal.

Main Contribution: The ICAN Method

- ICAN** (Information Capacity Approximate Network) is a novel method designed to estimate the confidence of a regression neural network per prediction.
- Distortion Assumption:** We model the model's predictor function using the following distortion and additive noise:
 $\hat{Y} = f(Y) + v$
- We show that the Shannon capacity can be used to analyze NN model predictions.
- Capacity via SNR: We estimate the capacity of a given prediction (sample) by estimating its Signal-to-Noise ratio (SNR).

In this form, we can use Shannon's capacity to estimate the maximum rate of our NN model at the sample's feature space vector X :

$$C = 0.5 \log_2(1 + \frac{Y^2}{\mathbb{E}\{v^2\}}) \text{ [bits/sec per channel usage]}$$

Per channel usage means **per inference**, in our case.

To estimate the capacity of the sample the model inferences, we need the estimate the SNR: $\text{SNR} = \frac{Y^2}{\mathbb{E}\{v^2\}}$

Noise Estimation (estimating $\mathbb{E}\{v^2\}$):

Take K samples from the validation set, whose features vector Euclidean distance from the sample's features vector is below a predefined threshold.

Calculate their model error, using the predictions error on the validation set:

$\mathbb{E}\{v^2\} = \mathbb{E}\{|\hat{Y} - Y|^2\}$ on this set **assuming statistical similarity**.

We calculate Shannon's capacity from the estimated SNR, assuming an AWGN channel. Samples are classified by their associated Shannon capacity value. We build ICAN-PI per **threshold_capacity = {1,2,3,4,...}**, as follows:

- Select all samples into the group T_i (associated with a **minimum capacity**) whose estimated capacity is **larger than the threshold capacity**.
- Calculate a PI with L_i and U_i (lower and higher bounds) in which 95% of the samples must fall within normal random variable, with:
 - mean = $\mathbb{E}\{T_i\}$
 - std_val = std_val $\{T_i\}$

Motivation: Classification by Capacity

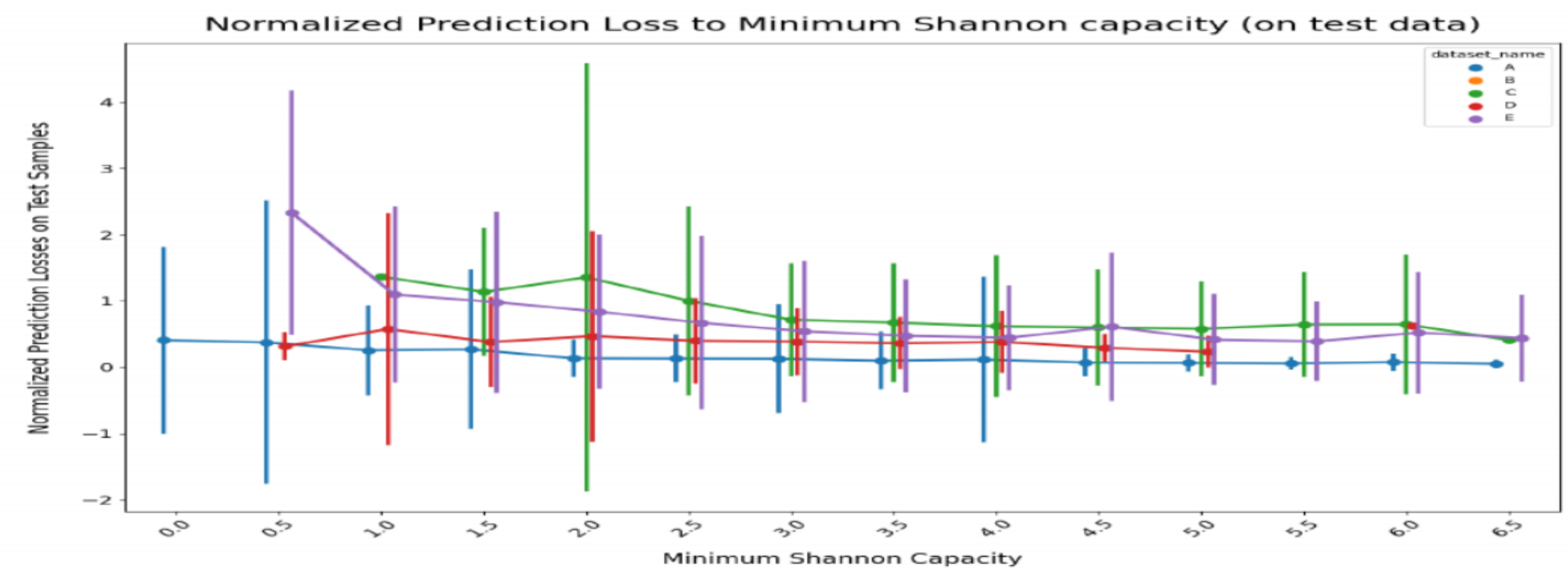


Fig. 1: Normalized loss as a function of capacity per dataset.

The Experiments

We tested ICAN on the following models:

- Approximate Query Processing (AQP) predictor of a SQL database query:
 - On 5 different datasets.
 - A set of Fully Connected models, trained on the following Kaggle datasets: Life expectancy dataset, Wine quality dataset, and Vehicle dataset.

We compared the results with multiple methods, such as: PIVEN, BNN.

Evaluation Metrics

We used the following evaluation metrics:

MPIW: Mean PI width:

$$\text{MPIW} := \frac{1}{n} \sum_{i=0}^n (U_i - L_i)$$

PICP: Prediction Interval Coverage Percentage:

$$\text{PICP} := \frac{1}{n} \sum_{i=0}^n k_i$$

Where, $k_i = 1$ if the i -th sample in the group T , falls between L_i and U_i .

The Results

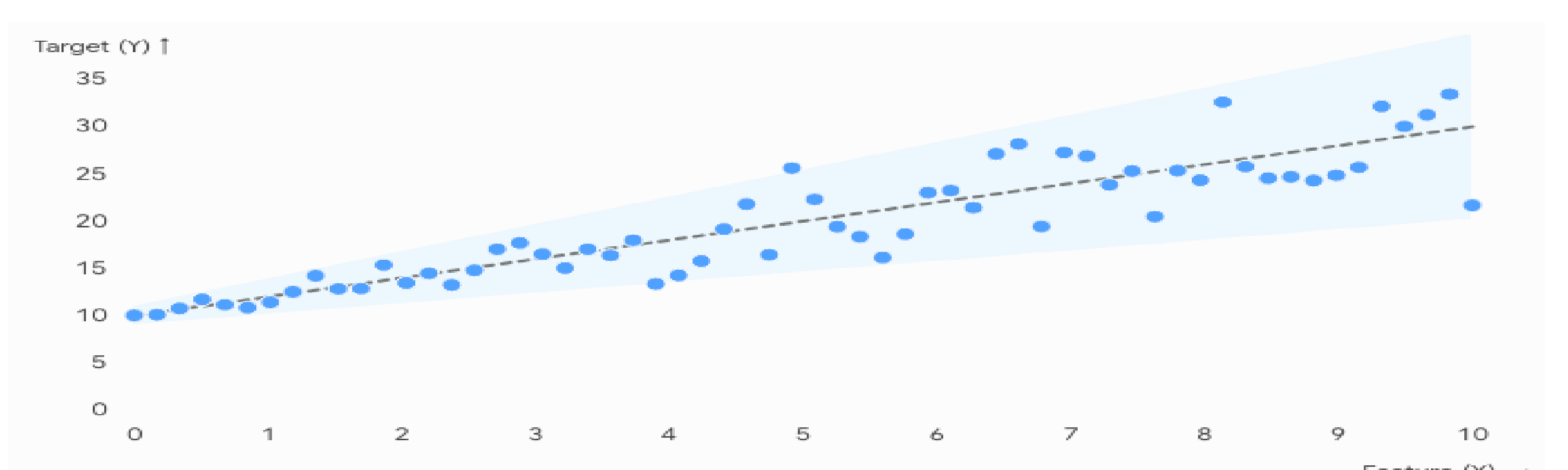
Table 1: Performance of the examined methods based on the PICP metric.

Dataset/Method	ICAN (Our Method)	Weighted-ICAN (Our Method)	PIVEN-over-AQP	BNN	PIVEN
A	0.49	0.48	0.45	NA	0.37
B	0.45	0.51	0.48	NA	0.36
C	0.55	0.55	0.58	NA	0.45
D	0.71	0.71	0.73	NA	0.15
E	0.66	0.65	0.45	NA	0.41
Life Exp.	0.95	0.96	NA	0.98	0.97
Wine(w)	1	1	NA	0.88	0.95
Wine(r)	0.98	0.98	NA	1	1
Vehicle	0.95	0.92	NA	1	0.91

Table 2: Performance of the examined methods based on the MPIW metric.

Dataset/Method	ICAN (Our Method)	Weighted-ICAN (Our Method)	PIVEN-over-AQP	BNN	PIVEN
A	5.4	5.4	5.5	NA	1.8
B	0.4	0.45	0.7	NA	1.6
C	2.8	2.75	2.7	NA	3.5
D	1.7	1.7	2.1	NA	2.6
E	1.1	1.1	1.6	NA	5.6
Life Exp.	0.6	0.61	NA	0.6	0.6
Wine(w)	0.1	0.1	NA	0.3	0.85
Wine(r)	0.2	0.21	NA	1	1.5
Vehicle	0.35	1.15	NA	1.15	1.45

Example: A visualization of a PI for a regression model



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